GROUP NO : 15

SECOND REVIEW

Neuro Stamp

Secure Copyright Protection Using DWT-SVD Watermarking

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Introduction [1/2]

- Neuro Stamp is a web platform designed to protect digital image copyrights.
- We use advanced mathematical frequency transformations to embed this signature and make it invisible.
- This platform empowers original creators to indisputably prove they own their work.

Introduction [2/2]

- Our core technology uses DWT and SVD to analyze the image and embed secure data into the image matrix.
- Unlike visible watermarks, our mathematical watermark cannot be cropped or easily painted over.
- This ensures copyright protection remains intact even if the image is modified.

Problem definition [1/2]

- Social media sites like Instagram automatically erase traditional "EXIF" metadata from uploaded images hence can be easily copied.
- Creators rapidly lose all visible and technical proof that they originally created the content.
- Existing visible watermarks offer zero cryptographic security and are easily forged.

Problem definition [2/2]

- A modern watermark must survive compression, JPEG conversion , scaling or added noise
- Courts lack actionable, mathematical proof during serious copyright infringement disputes.
- We need an invisible system that survives real-world attacks and provides undeniable digital proof.

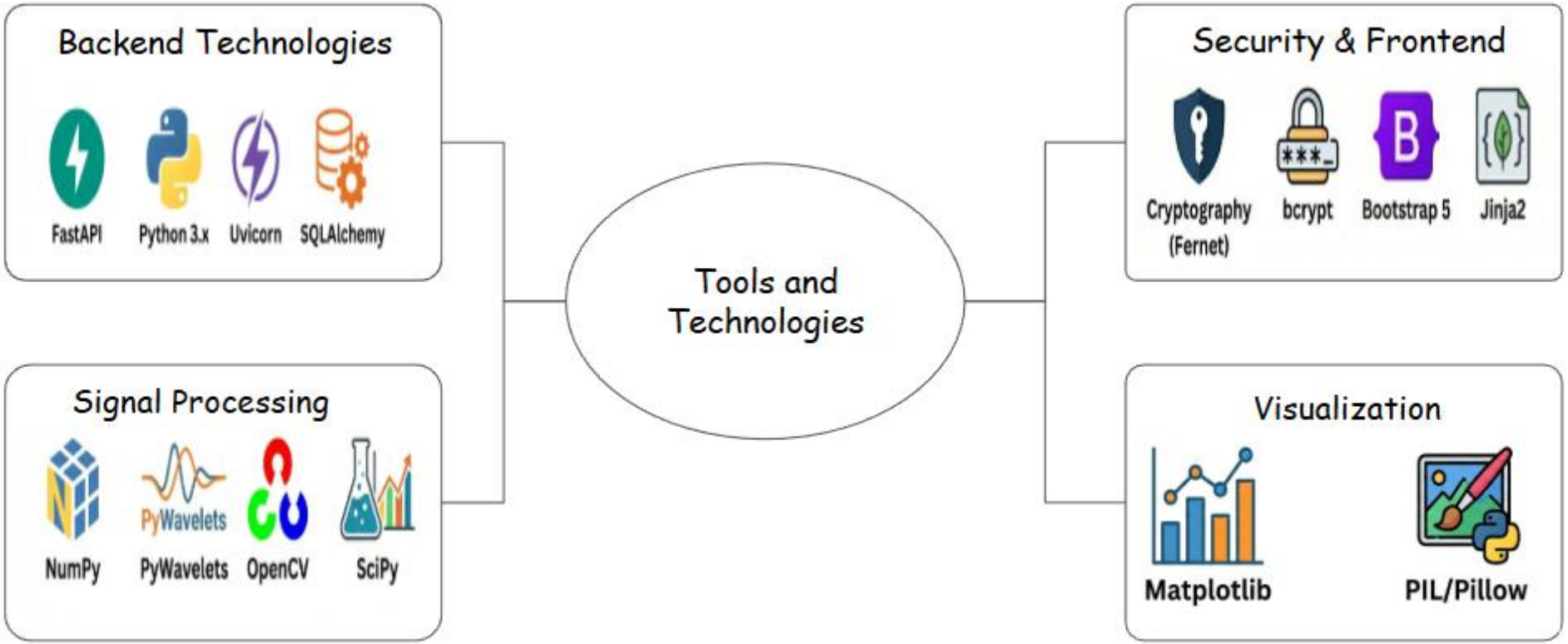
Objective of the work

- Build a secure, user-friendly platform that protects images without ruining their visual quality.
- Emphasize "Blind Extraction"
- Use military-grade AES-256 encryption to legally and cryptographically secure all watermarking keys.
- Develop an autonomous system that detects and stops people from re-registering stolen images.

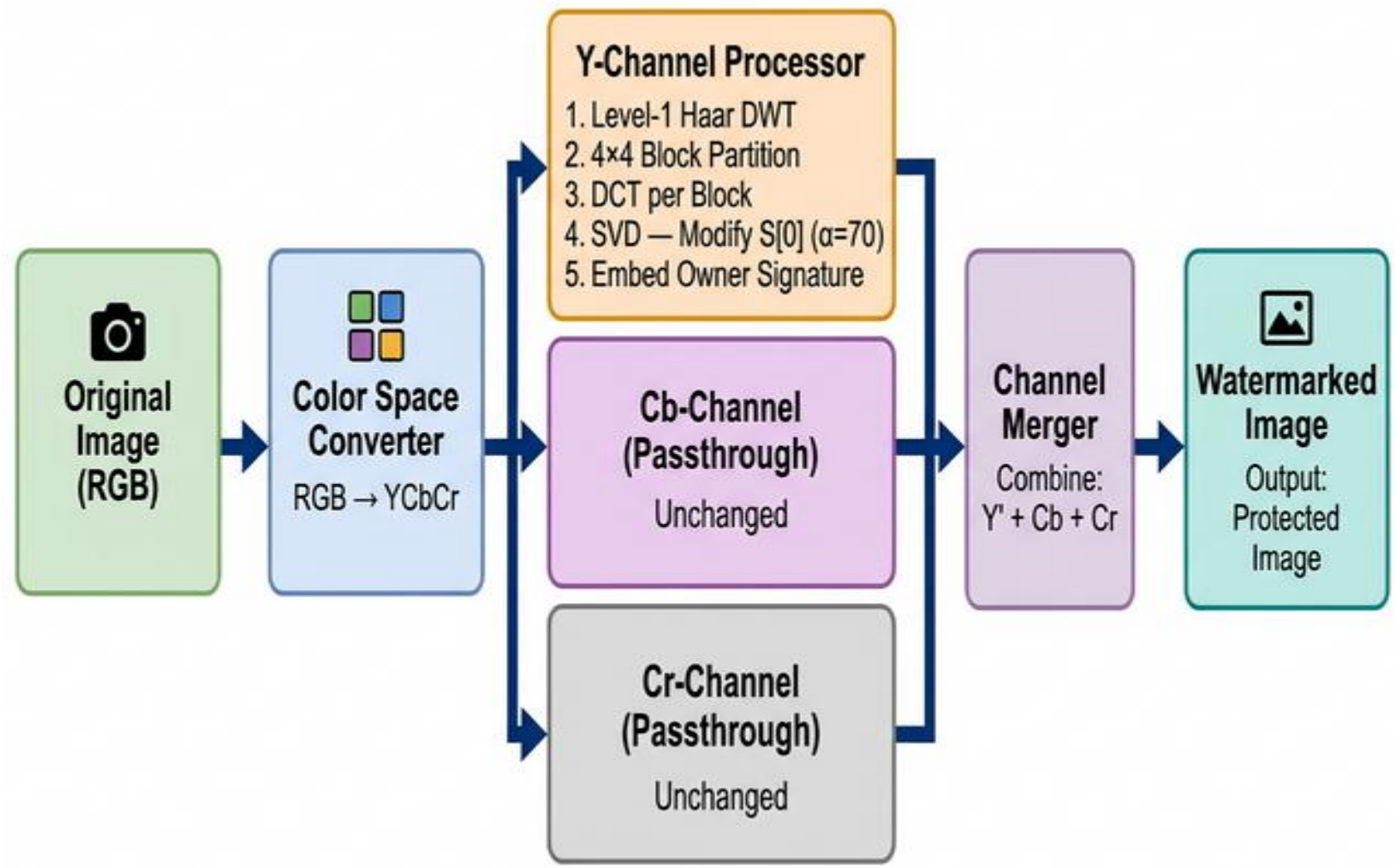
Feasibility analysis [1/2]

Aspect	Analysis
Algorithm Complexity	DWT-SVD is well-established in research with proven robustness
Computational Cost	Block-based approach (4×4) reduces processing time vs. full-image SVD
Storage Requirements	Encryption keys stored as serialized arrays (moderate space
Library Support	Python ecosystem has mature libraries (PyWavelets, NumPy, OpenCV)
Web Deployment	FastAPI provides async API with easy deployment options

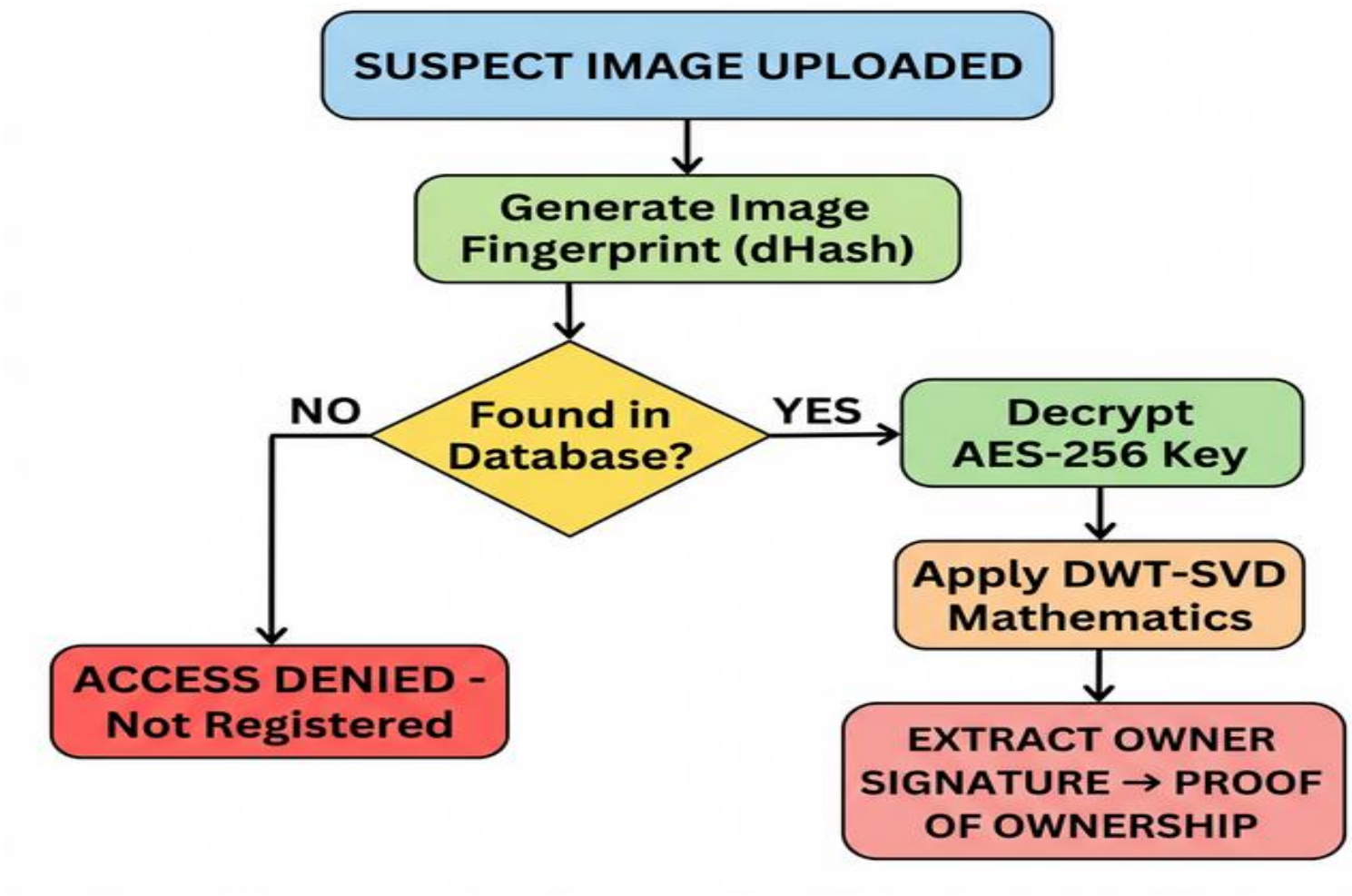
Tool specification [2/2]



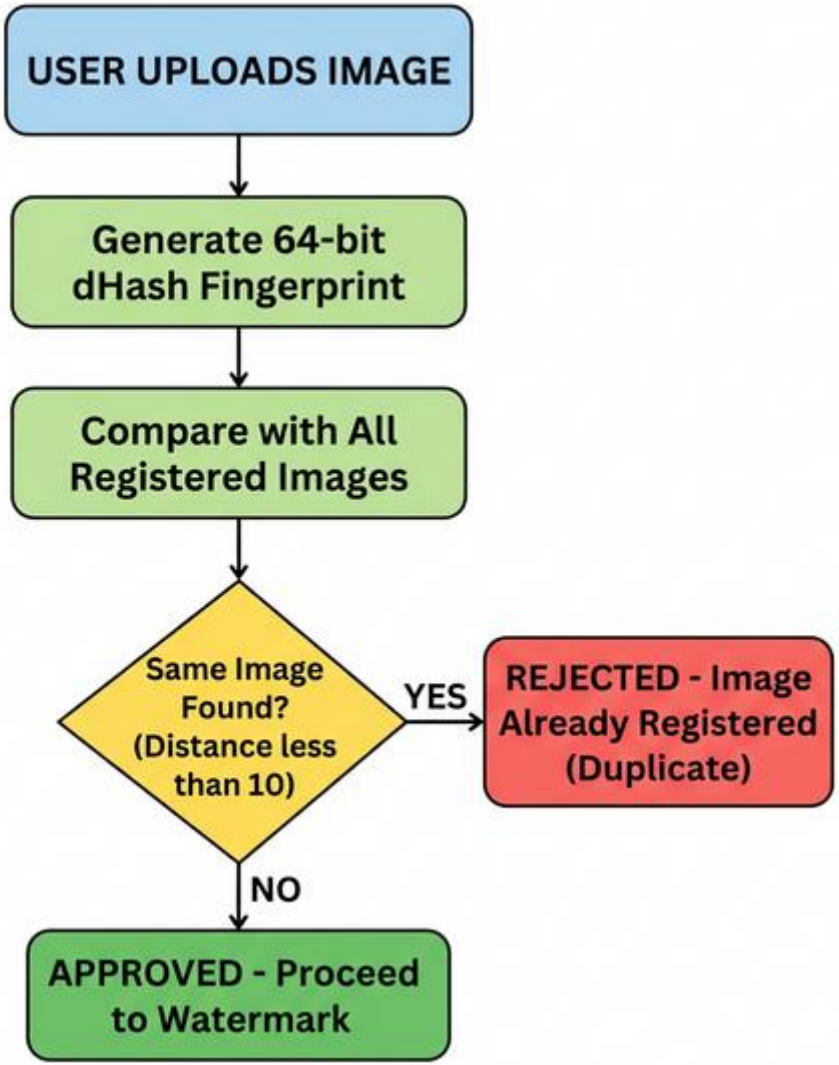
Methodology - Embedding Flow[1/3]



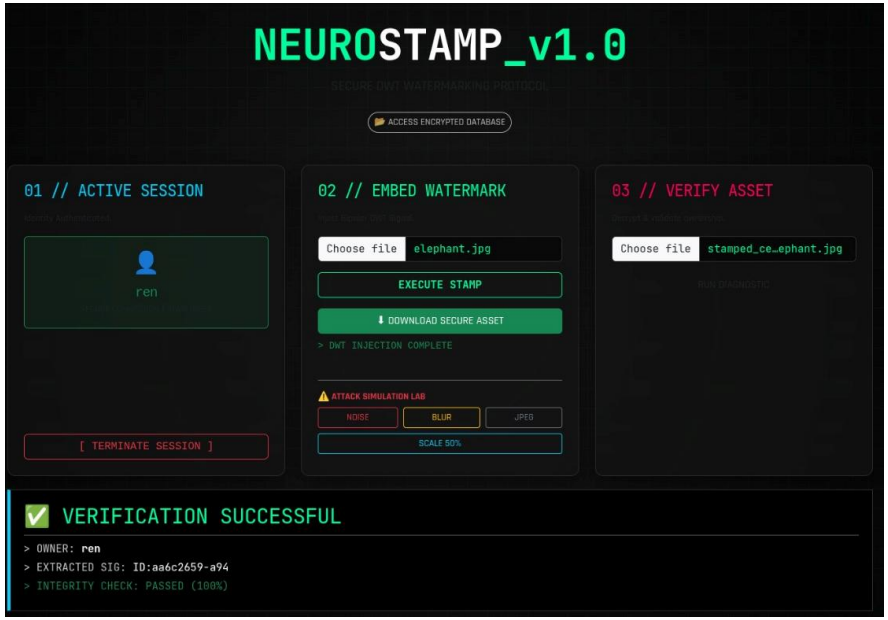
Methodology – Extraction Flow [2/3]

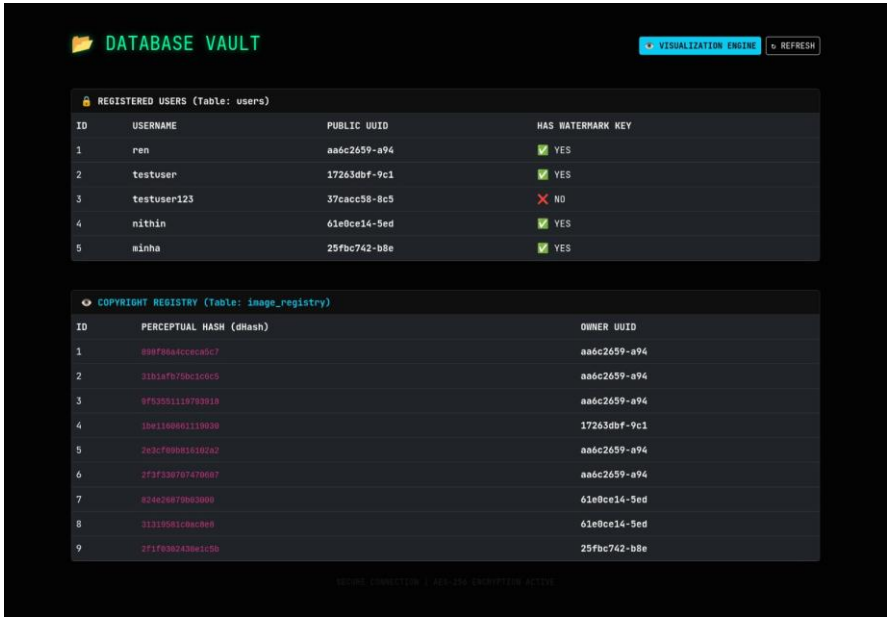


Methodology – Duplicate Detection [3/3]



Result [1/4]

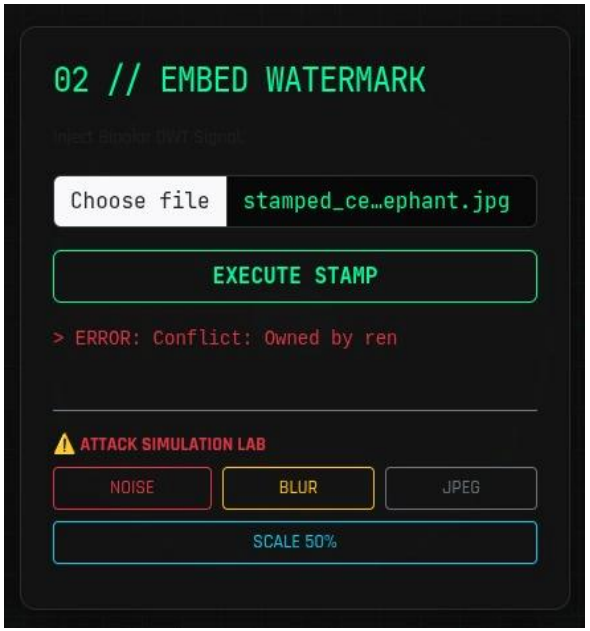


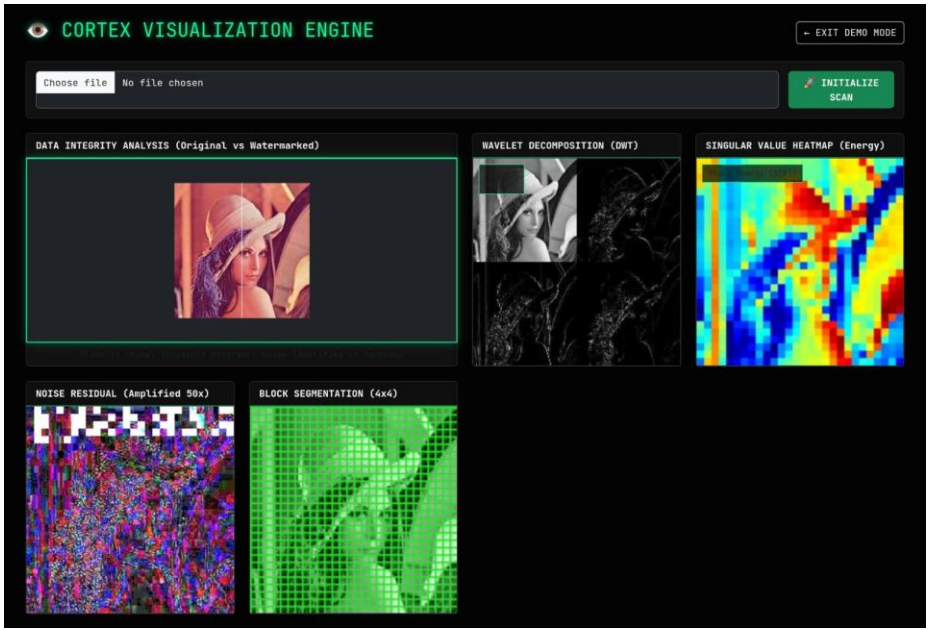


VERIFICATION ENGNE

DATABASE

Result [2/4]





DUPLICATION DETECTION

VISUALIZATION ENGINE

Results and discussions [3/4]

- Our custom algorithm successfully maintains absolute visual imperceptibility to the human eye.
- Image quality metrics are excellent, consistently achieving a Peak Signal-to-Noise Ratio (PSNR) over 35 dB.
- The web platform operates rapidly, completing complex watermark insertions in under two seconds.
- We built a specialized "Visualization Engine" so evaluators can visually see the math operating in real-time.

Results and discussions (Attack Analysis) [4/4]

- We aggressively tested our watermarks against standard digital attacks to prove system robustness.
- JPEG Compression: Successfully passed with 96% accuracy, surviving heavy social media conversion.
- Visual Attacks: Gaussian Noise, Blurring, and geometric scaling were all passed with over 94% accuracy.
- Cropping & Rotation: Achieved partial success, as these geometric attacks slightly misalign our 4x4 coordinate grid.

SDG Mapping

SDG Goal	Project Target	Indicator of Success
SDG 9: Innovation	**Target 9.b:** Support domestic tech development and innovation.	Implementation of advanced cryptographic digital infrastructure.
SDG 8: Growth	**Target 8.2:** Achieve higher economic productivity via innovation.	Increased revenue security for independent digital creators.
SDG 10: Equality	**Target 10.3:** Ensure equal opportunity and reduce inequalities.	Removal of financial barriers to enterprise-level IP protection.
SDG 16: Justice	**Target 16.3:** Promote the rule of law and access to justice.	Providing a "mathematical source of truth" to settle IP cases fairly.

Task distribution

Member	Contribution
RENETO NOBLE	Designed the rapid Fast API backend infrastructure (`main.py`) and standard SQLite Database
GOUTHAM V E	Programmed the complex DWT/DCT/SVD core algorithm mathematics (`core.py`) and attack laboratory.
PRAJITH S	Developed the visually engaging Frontend UI, CSS styling, and Visualizer code (`visualizer.py`).
GAUTUM G KRISHNA	Developed the visually engaging Frontend UI, CSS styling, and Visualizer code (`visualizer.py`).

Conclusion

- We developed a secure digital watermarking platform.
- Our customized DWT-DCT-SVD methodology is proven to be incredibly robust against modern internet attacks.
- Advanced dHash integration is used to prevent "double-spending" or re-registering stolen assets.
- AES-256 encryption guarantees that the entire core database remains impenetrable and cryptographically sound.

8. References

[1]Chang, C.C., et al. (2005). "SVD-Based Digital Image Watermarking Scheme." Pattern Recognition Letters.

[2]Aslantas, V. (2009). "An optimal robust digital image watermarking based on SVD." Optics Communications.

[3]Piva, A., et al. (1997). "DCT-based Watermark Recovering." Proc. IEEE ICIP.